# CP2.8 — Modelling with Differential Equations

## 1. Simple Harmonic Motion (SHM)

**SHM equation:**

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

This is the homogeneous ODE with AE $r^2 + \omega^2 = 0$, giving complex roots $r = \pm\omega i$.

### 1.1 Solution and Forms

$$x(t) = A\cos\omega t + B\sin\omega t$$

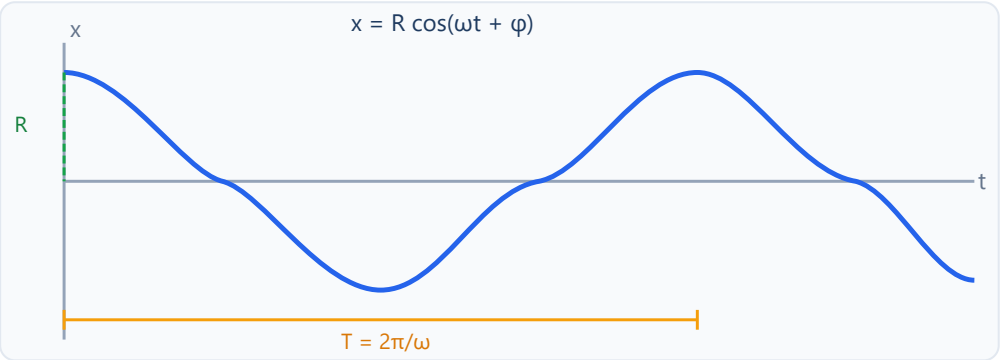
This can also be written in amplitude-phase form:

$$x(t) = R\cos(\omega t + \phi)$$

where $R = \sqrt{A^2 + B^2}$ (amplitude), and $\tan\phi = -B/A$ (or use $R\cos\phi = A$, $R\sin\phi = -B$ — check carefully).

| Quantity | Formula |
|---|---|
| Angular frequency | $\omega$ (rad/s) |
| Period | $T = 2\pi/\omega$ |
| Frequency | $f = 1/T = \omega/(2\pi)$ |

| | |
|---|---|
| Amplitude | $R = \sqrt{A^2 + B^2}$ |
| Velocity | $\dot{x} = -R\omega \sin(\omega t + \phi)$ |
| Max speed | $R\omega$ (when $x = 0$) |
| Max acceleration | $R\omega^2$ (when $x = \pm R$) |




**Example:** A particle moves with SHM such that $\ddot{x} = -16x$. At $t = 0$: $x = 3$, $\dot{x} = 0$.

$\omega = 4$. GS: $x = A\cos 4t + B\sin 4t$.
$x(0) = 3 \Rightarrow A = 3$.
$\dot{x} = -4A\sin 4t + 4B\cos 4t$. $\dot{x}(0) = 0 \Rightarrow 4B = 0 \Rightarrow B = 0$.
Solution: $x = 3\cos 4t$. Amplitude $= 3$, period $= \pi/2$.

# 2. Damped Oscillations

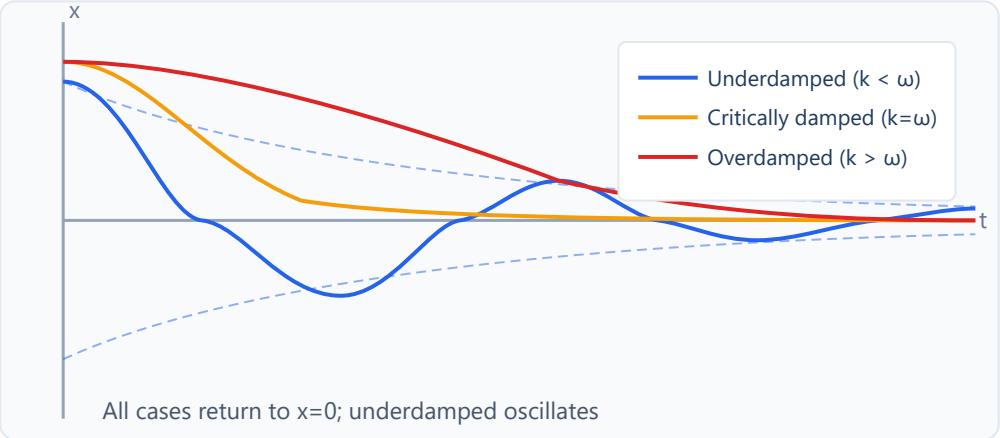
The **damped oscillation equation**:

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + \omega^2 x = 0$$

Auxiliary equation: $r^2 + 2kr + \omega^2 = 0$, so $r = -k \pm \sqrt{k^2 - \omega^2}$.

## 2.1 Three Cases

| Case | Condition | Roots | Solution form | Behaviour |
|---|---|---|---|---|
| Overdamped | $k > \omega$ | Two distinct negative reals | $x = Ae^{r_1 t} + Be^{r_2 t}$ | Returns to equilibrium without oscillating, slowly |
| Critically damped | $k = \omega$ | Repeated real $r = -k$ | $x = (A + Bt)e^{-kt}$ | Fastest return to equilibrium without oscillating |
| Underdamped | $k < \omega$ | Complex $-k \pm \mu i$, $\mu = \sqrt{\omega^2 - k^2}$ | $x = e^{-kt}(A\cos\mu t + B\sin\mu t)$ | Oscillates with decreasing amplitude |




**Example (Underdamped):** Solve $\ddot{x} + 4\dot{x} + 13x = 0$, $x(0) = 2$, $\dot{x}(0) = 0$.

AE: $r^2 + 4r + 13 = 0 \Rightarrow r = -2 \pm 3i$. $k = 2$, $\mu = 3$.
GS: $x = e^{-2t}(A\cos 3t + B\sin 3t)$.
$x(0) = 2 \Rightarrow A = 2$.
$\dot{x} = e^{-2t}[(-2A + 3B)\cos 3t + (-3A - 2B)\sin 3t]$ Wait — let me be careful: $\dot{x}(0) = -2A + 3B = 0 \Rightarrow B = 4/3$.
Solution: $x = e^{-2t}\left(2\cos 3t + \frac{4}{3}\sin 3t\right)$.

## 3. Forced Oscillations and Resonance

The **forced oscillation equation**:

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + \omega^2 x = F\cos\Omega t$$

The **particular integral** gives the **steady-state response**; the CF gives the **transient**.

**Resonance** occurs when the forcing frequency $\Omega$ equals the natural frequency $\omega$ of the system (and there is no damping). The amplitude grows without bound.

**Undamped resonance:** $\ddot{x} + \omega^2 x = F\cos\omega t$.
The trial PI $a\cos\omega t + b\sin\omega t$ is in the CF — use $x_p = t(a\cos\omega t + b\sin\omega t)$.
This gives $x_p = \dfrac{F}{2\omega} t\sin\omega t$ — amplitude grows linearly with time.

**Example:** Solve $\ddot{x} + 4x = 3\cos 2t$, $x(0) = 0$, $\dot{x}(0) = 0$.

CF: $A\cos 2t + B\sin 2t$. Resonance: try $x_p = t(a\cos 2t + b\sin 2t)$.
$\dot{x}_p = a\cos 2t + b\sin 2t + t(-2a\sin 2t + 2b\cos 2t)$
$\ddot{x}_p = -4a\sin 2t + 4b\cos 2t + t(-4a\cos 2t - 4b\sin 2t)$
$\ddot{x}_p + 4x_p = -4a\sin 2t + 4b\cos 2t = 3\cos 2t$
$b = 3/4$, $a = 0$. PI: $x_p = \dfrac{3t\sin 2t}{4}$.
GS: $x = A\cos 2t + B\sin 2t + \dfrac{3t\sin 2t}{4}$.
$x(0) = 0 \Rightarrow A = 0$. $\dot{x}(0) = 2B + 0 = 0 \Rightarrow B = 0$.
Solution: $x = \dfrac{3t\sin 2t}{4}$. Amplitude $\dfrac{3t}{4}$ grows unboundedly.

## 4. Coupled Differential Equations

Two simultaneous ODEs in two unknowns $x(t)$ and $y(t)$ can be solved by **elimination**: differentiate one equation and substitute the other to get a single second-order ODE in one variable.

**Example:** Solve the coupled system:

$$\frac{dx}{dt} = 3x - 2y, \quad \frac{dy}{dt} = 2x - y$$

Differentiate the first: $\ddot{x} = 3\dot{x} - 2\dot{y}$.
From the second: $\dot{y} = 2x - y$. Substitute: $\ddot{x} = 3\dot{x} - 2(2x - y) = 3\dot{x} - 4x + 2y$.
From the first: $2y = 3x - \dot{x}$, so $y = \frac{3x - \dot{x}}{2}$. Substitute:
$\ddot{x} = 3\dot{x} - 4x + 3x - \dot{x} = 2\dot{x} - x$
$\ddot{x} - 2\dot{x} + x = 0$. AE: $(r-1)^2 = 0$, $r = 1$ (repeated).
$x = (A + Bt)e^t$. Then $y = \frac{3x - \dot{x}}{2}$:
$\dot{x} = (A + Bt)e^t + Be^t = (A + B + Bt)e^t$.
$y = \frac{(3A + 3Bt - A - B - Bt)e^t}{2} = \frac{(2A - B + 2Bt)e^t}{2} = \left(A - \frac{B}{2} + Bt\right)e^t$.

**Exam tip:** For coupled ODEs, the method is: differentiate one equation, express the eliminated variable from the other, and substitute. Expect a 2nd-order ODE. Once $x(t)$ is found, find $y(t)$ algebraically from one of the original equations.

## 5. Predator-Prey and Population Models

Coupled ODEs model interacting populations. The **Lotka-Volterra** system is non-linear (beyond the spec), but linear approximations near equilibria are examinable.

**Example:** In a model, prey $x$ and predator $y$ satisfy:

$$\frac{dx}{dt} = ax - bxy, \quad \frac{dy}{dt} = cxy - dy$$

Near equilibrium $(\bar{x}, \bar{y}) = (d/c, a/b)$, linearised equations are coupled first-order ODEs — solve as above.

## 6. Other Modelling Scenarios

## 6.1 Population Growth — Logistic Equation

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

where $r$ = growth rate, $K$ = carrying capacity. This is a separable first-order ODE.

**Solution:** Using partial fractions on $\int \frac{dP}{P(1-P/K)}$:

$$P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0}\right) e^{-rt}}$$

As $t \to \infty$, $P \to K$ (carrying capacity). The solution is the logistic curve — S-shaped.

## 6.2 Newton's Law of Cooling

$$\frac{d\theta}{dt} = -k(\theta - \theta_{\text{amb}})$$

where $\theta$ = temperature of object, $\theta_{\text{amb}}$ = ambient temperature, $k > 0$.
This is a first-order linear ODE. Solution: $\theta = \theta_{\text{amb}} + (\theta_0 - \theta_{\text{amb}})e^{-kt}$.

## 6.3 Formulating a DE from Context

Key steps:

1. Identify the dependent variable and what it depends on
2. Write a word equation: "rate of change = inflow - outflow" (or similar)
3. Translate to mathematical symbols
4. Identify the type of ODE and solve

**Example:** A tank contains 1000 L of water with 50 kg of salt dissolved. Pure water enters at 10 L/min and the well-mixed solution leaves at 10 L/min. Find the amount of salt $S(t)$ at time $t$.

Rate in = $0$ kg/min (pure water). Rate out = $\frac{S}{1000} \cdot 10 = \frac{S}{100}$ kg/min.
ODE: $\frac{dS}{dt} = -\frac{S}{100}$. This separates: $S = 50e^{-t/100}$.

## 7. SHM — Worked Energy Connection

From $\ddot{x} = -\omega^2 x$, writing $\ddot{x} = v\frac{dv}{dx}$:

$$v\frac{dv}{dx} = -\omega^2 x \Rightarrow \frac{v^2}{2} = \frac{\omega^2(R^2 - x^2)}{2}$$

$$v^2 = \omega^2(R^2 - x^2)$$

Maximum speed $= \omega R$ at $x = 0$; zero speed at $x = \pm R$ (amplitude).

## 8. Summary Table — Modelling Equations

| Model | ODE | Solution form |
|---|---|---|
| SHM | $\ddot{x} = -\omega^2 x$ | $R\cos(\omega t + \phi)$ |
| Overdamped | $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$, $k > \omega$ | $Ae^{r_1 t} + Be^{r_2 t}$ (both $r < 0$) |
| Critical damping | Same, $k = \omega$ | $(A + Bt)e^{-kt}$ |
| Underdamped | Same, $k < \omega$ | $e^{-kt}(A\cos\mu t + B\sin\mu t)$ |
| Resonance | $\ddot{x} + \omega^2 x = F\cos\omega t$ | Amplitude $\propto t$ |
| Newton's cooling | $\dot{\theta} = -k(\theta - \theta_a)$ | $\theta_a + Ce^{-kt}$ |
| Logistic growth | $\dot{P} = rP(1 - P/K)$ | S-shaped, $P \to K$ |

# 9. Exam-Style Practice Questions

1. **[SHM]** A particle oscillates with SHM described by $\ddot{x} + 25x = 0$. Given $x = 4$ and $\dot{x} = -15$ when $t = 0$:
   (a) Find $x$ as a function of $t$.
   (b) Find the amplitude and period.
   (c) Write $x$ in the form $R\cos(5t + \phi)$.
2. **[Damping]** Solve $\ddot{x} + 6\dot{x} + 9x = 0$ with $x(0) = 1$, $\dot{x}(0) = 2$. Identify the type of damping. Sketch the solution.
3. **[Forced oscillation]** Find the general solution of $\ddot{x} + 9x = 12\cos 3t$. Comment on the physical significance.
4. **[Coupled ODEs]** Solve the system $\dot{x} = 2x + y$, $\dot{y} = x + 2y$ with $x(0) = 3$, $y(0) = 1$.
5. **[Modelling]** The temperature $\theta$°C of a metal bar satisfies Newton's law of cooling with ambient temperature 20°C. Initially $\theta = 80$°C and after 5 minutes $\theta = 60$°C. Find (a) the value of $k$, (b) the time when $\theta = 30$°C.
6. **[Logistic]** A population grows logistically with $r = 0.1$ per year and $K = 5000$. Initially $P = 500$. Find the population after 10 years.

**General exam tips for CP2.8:**

- SHM: state the ODE in the form $\ddot{x} = -\omega^2 x$ explicitly, then quote the general solution form
- Damping: always identify which case (over/critical/under) before solving
- Forced resonance: if $\Omega = \omega$ appears, immediately flag that the standard PI fails and multiply by $t$
- Coupled ODEs: eliminate one variable completely — do not try to integrate a coupled system directly
- Modelling: define your variables carefully; identify the type of ODE before solving
- Check units and physical sense in the answer (e.g., population cannot be negative)

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